

15 July 24:-

Referred book $\rightarrow$ {E. Balagurusamy}

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Numerical Methods {Applied Mathematics}

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Algebraic equations:-

$$(ax^2 + bx + c)$$

$x = ?$

Quadratic eqn.

$$[ax^n + bx^{n-1} + \dots + zx^0 = 0]$$

Polynomial equations:-

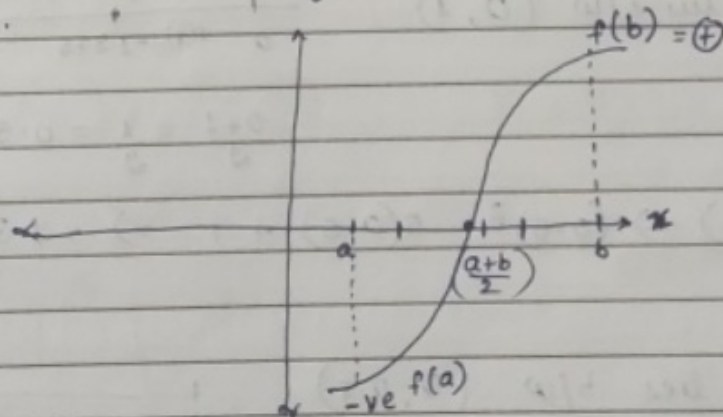
$\Rightarrow$ Transcendental equations:-

$\log x$, $\sin x$, exponential

$$[\log x - e^x = \sin y]$$

★ Bisection Method (Bolzano Method):- 1st Method

If $f(x)$ {it is a func. in terms of x } is continuous in an interval (a, b) such that $f(a) = -ve$ and $f(b)$ is positive, then we say root of function lies b/w a & b .



$$y^2 = 4ax \rightarrow \text{Parabolic eqn.}$$

$$4ax = 0$$

{line}

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Ques $\rightarrow$ (1) $[x^2 + 2x - 8 = 0]$

Solⁿ:- (1)

$$f(0) = -8$$

$$f(1) = 1 + 2 - 8 = -5$$

$$f(2) = 4 + 4 - 8 = 0$$

$$f(3) = 9 + 6 - 8 = 7$$

$\rightarrow$ Root lies b/w (1, 3)

$$\begin{array}{c} 1 \quad \quad \quad 3 \\ | \quad \quad \quad | \\ 1 \quad \quad \quad 2 \quad f(x)=0 \quad 3 \end{array}$$

$$\frac{1+3}{2} \Rightarrow \frac{4}{2} = 2$$

$$f(2) = 0$$

Ques $\rightarrow$ (2) $x^3 - 5x + 1 = 0$

Solⁿ:- (2)

$$f(0) = 1$$

$$f(1) = -3$$

$\rightarrow$ Root lies b/w (0, 1)

$$\begin{array}{c} 0.5 \\ | \\ 0 \quad f(x) = -1.375 \quad 1 \end{array}$$

$$\frac{0+1}{2} = \frac{1}{2} = 0.5$$

$$f(0.5) = (0.5)^3 - 5(0.5) + 1 = -1.375$$

$\rightarrow$ Root lies b/w (0, 0.5)

$$\begin{array}{c} 0.25 \\ | \\ 0 \quad \quad \quad 0.5 \end{array}$$

$$\frac{0+0.5}{2} = \frac{0.5}{2} = 0.25$$

$$f(0.25) = (0.25)^3 - 5(0.25) + 1 = -0.234$$

→ Root lies b/w (0, 0.25) $\Rightarrow 0.125$

$$f(0.125) = (0.125)^3 - 5(0.125) + 1 = 0.000$$

$$(0.125) \Rightarrow 0.376$$

→ Root lies b/w (0.25, 0.125) $\Rightarrow \frac{0.25+0.125}{2} = 0.1875$

$$(0.1875)^3 - 5(0.1875) + 1 = 0.0690$$

{ Note :- up to three decimal digit comes same $\rightarrow$ stop there }

Ques → (3) $x^3 - 2x - 5 = 0$

$$f(0) = -5$$

$$f(1) = -4$$

$$f(2) = -1$$

$$f(3) = 16$$

→ Root lies b/w (2, 3) $x_1 = \frac{2+3}{2} = 2.5$

$$\Rightarrow (2.5)^3 - 2(2.5) - 5 = 0 \Rightarrow f(2.5)$$

$$5.625 = f(x)$$

→ Root lies b/w (2, 2.5) $\Rightarrow \frac{2+2.5}{2} = 2.25$

$$(2.25)^3 - 2(2.25) - 5 = 0$$

$$1.89$$

→ Root lies b/w (2 & 2.25) $x_3 = \frac{2 + 2.5}{2}$

$$\Rightarrow 2.125$$

$$(2.125)^3 - 2 \times (2.125) - 5 \Rightarrow f(2.125)$$

$$f(2.125) = 0.345$$

→ Root lies b/w (2 & 2.125)

$$x_4 = \frac{2 + 2.125}{2} \Rightarrow 2.0625$$

$$f(2.0625) = (2.0625)^3 - 2 \times (2.0625) - 5$$

$$= -0.351$$

→ Root lies b/w (2.125 & 2.0625)

$$x_5 = 2.09375$$

$$(2.09375)^3 - 2(2.09375) - 5 = f(2.09375)$$

$$= -0.008941$$

→ Root lies b/w (2.125 & 2.09375)

$$x_6 = 2.109375$$

$$f(2.109375) \Rightarrow (2.109375)^3 - 2(2.109375) - 5$$

$$\Rightarrow 0.1668$$

→ Root lies b/w (2.09375 & 2.109375)

$$x_7 = 2.09766$$

$$f(2.09766) \Rightarrow 0.03471$$

→ Root lies b/w (2.09375 & 2.09766)

$$x_8 = 2.09570$$

$$f(x) = > 0$$

→ Root lies b/w (2.09375 & 2.09570)

$$x_9 = 2.09473$$

$$f(x) = > 0$$

→ Root lies b/w (2.09375 & 2.09473)

$$x_{10} = \boxed{2.09425} \rightarrow \text{(Ans)}$$

$$f(x) = < 0$$

→ Root lies b/w (2.09424 & 2.09473)

Same upto 3 decimals

Q.No.	⊖	⊕	$\frac{a+b}{2}$	$f(x)$
1	2	3	2.5	5.625 > 0
2	2	2.5	2.25	> 0
3	2	2.25	2.125	> 0
4	2	2.125	2.0625	-0.351
5	2.0625	2.125	2.0625	-0.0089 < 0
6	2.09375	2.125	2.09375	0.1668 > 0
7	2.09375	2.109375	2.109375	0.3471 > 0
8	2.09375	2.09766	2.09766	> 0
9	2.09375	2.09570	2.09570	> 0
10	2.09375	2.09473	2.09473	< 0
11	2.09424	2.09473	2.09424	

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{ Rate of convergence = 0.5 } Date: / / Page:

Q → $x - \cos x = 0$

Q → $f(x) = x - \cos x$

$f(1) = 1 - 0.99 = 0.01 > 0$

$f(0) = -1 < 0$

→ Hence, the root lies b/w -1 & 0

$x_1 = 0.5$

$f(0.5) = -0.3775 < 0$

→ The

~~$x_2 = \frac{0.3775 + 0.4596}{2} = 0.4185$~~

$x_2 = \frac{0.5 + 1}{2} = \frac{1.5}{2} = 0.75$

$f(0.75) = 0.0183 > 0$

→ $x_3 = \frac{0.75 + 0.5}{2} = 0.625$

$f(0.625) = -0.1859 < 0$

→ $x_4 = \frac{0.625 + 0.75}{2} = 0.6875$

$f(0.6875) = -0.0853$

→ $x_5 = \frac{0.6875 + 0.75}{2} = 0.71875$

$f(0.71875) = -0.0338 < 0$

$$\rightarrow X_6 = \frac{0.71875 + 0.75}{2} = 0.734375$$

$$f(0.734375) = -0.0018$$

$$\rightarrow X_7 = \frac{0.734375 + 0.75}{2} = 0.7421875$$

$$f(0.7421875) = 0.00519$$

$$\rightarrow X_8 = \frac{0.7421875 + 0.734375}{2} = 0.73828125$$

$$\Rightarrow f(0.73828125) = -0.001345$$

$$Q \rightarrow x^3 - 5x - 7 = 0$$

$$f(0) = -7$$

$$f(1) = -11$$

$$f(2) = -9$$

$$f(3) = 5$$

$$\rightarrow X_1 = \frac{2+3}{2} = 2.5$$

$$f(2.5) = -9.875$$

$$\rightarrow X_2 = \frac{2.5 + 3}{2} = 2.75$$

$$f(2.75) = 0.0468$$

$$\rightarrow X_3 = \frac{2.75 + 2.5}{2} = 2.625$$

$$f(2.625) = -2.0371$$

$$\rightarrow X_5 = \frac{2.625 + 2.75}{2} = 2.6875$$

$$f(2.6875) = -2.9860$$

$$\rightarrow X_6 = \frac{2.6875 + 2.75}{2} = 2.71875$$

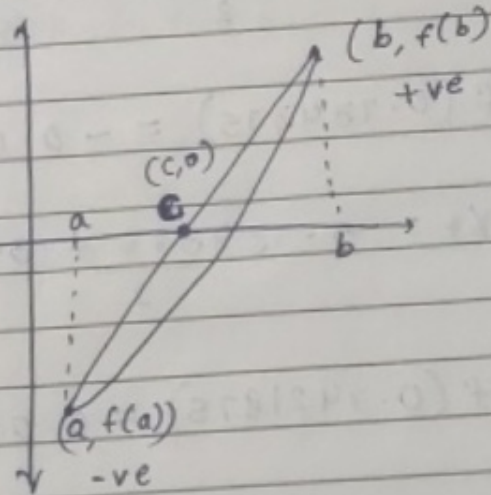
$$\Rightarrow f(x) = -1.5396$$

IInd Method:-

Regula-Falsi Method :-

False Position Method

Rate of conversion $\rightarrow 1.618$



$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{slope of AC} = \text{slope of AB}$$

$$\frac{0 - f(a)}{c - a} = \frac{f(b) - f(a)}{b - a}$$

$$c = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

$$\rightarrow f(a) \cdot f(c) < 0 \quad (a, c)$$

$$\rightarrow f(b) \cdot f(c) < 0 \quad (b, c)$$

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→ Regula Falsi method :-

Q → $x^3 - 5x - 7 = 0$ { Algebraic / Polynomial eqn. }

$$f(x) = x^3 - 5x - 7$$

$$f(0) = -7$$

$$f(1) = -11$$

$$f(2) = -9$$

$$f(3) = 5$$

→ Hence, the root lies b/w 2, 3

$$C_1 = \frac{2(5) - 3(-9)}{5 + 9} \Rightarrow \frac{10 + 27}{14} = \frac{37}{14} = 2.6428$$

$$f(2.6428) = -1.7556$$

$$\rightarrow C_2 = \frac{18.4808}{6.7556} = 2.7356$$

$$f(2.7356) = -0.2061$$

$$\rightarrow C_3 = \frac{14.2963}{5.2061} = 2.7460$$

$$f(2.7460) = -0.0237$$

$$\rightarrow C_4 = \frac{13.8011}{5.0237} = 2.7471, \quad f(2.7471) = -0.0043$$

$$\rightarrow C_5 = \frac{13.7484}{5.0043} = 2.7473, \quad f(2.7473) = -0.0043$$

a	b	c	f(c)
2	3	2.6428	-1.7556
2.6428	3	2.7356	-0.2061
2.7356	3	2.7460	-0.0237
2.7460	3	2.7471	-0.0043
2.7471	3	2.7473	-0.0043

root $\rightarrow$ 2.7471 // Ans.

Q $\rightarrow$ $f(x) = x \log_{10} x - 1.2$ { Transcendental eqn. }

$$f(0) = -1.2$$

$$f(1) = 0 - 1.2 = -1.2$$

$$f(2) = 0.3010 - 1.2 = -0.8989$$

$$f(3) = 0.4771 - 1.2 = -0.7229$$

$$f(2.5) = 0.3979 - 1.2 = -0.8021$$

$\rightarrow$ Hence, the root lies b/w (2.5) & 3

$$c_1 = \frac{1.19355}{0.4364} \Rightarrow 2.7348, \quad f(2.7348) = -0.0050$$

$$\Rightarrow c_2 \Rightarrow \frac{0.6475}{0.2363} \Rightarrow 2.7401, \quad f(2.7401) = -0.0009$$

$$\rightarrow c_3 = 2.7406, \quad f(2.7406) = -0.000040$$

a	b	c	f(c)
2.5	3	2.7348	-0.0050
2.7348	3	2.74041	-0.00021
2.74041	3	2.7406	-0.000040
2.7406	3	2.7406	-0.000040
root $\rightarrow$ <u>2.7406</u>			

$$C_4 \Rightarrow \frac{2.7406 (0.2313) - 3 (-0.000040)}{0.2313 - (-0.000040)}$$

$$C_4 \Rightarrow \frac{0.6339 + 0.00012}{0.23134} \Rightarrow \frac{0.63402}{0.23134} = 2.7406$$

$$f(2.7406) = -0.000040$$

$$Q \rightarrow f(x) = x^3 + 2x^2 + 10x - 20$$

$$f(0) = -20$$

$$f(1) = -7$$

$$f(2) = 16$$

$$f(1.5) = 2.875$$

$$\rightarrow C_1 = \frac{13.375}{9.875} = 1.3544, f(1.3544) = -0.3026$$

$$\rightarrow C_2 = \frac{4.3478}{3.1776} = 1.3682, f(1.3682) = -0.0128$$

$$\rightarrow C_3 = \frac{3.9527}{2.8878} = 1.3687, f(1.3687) = -0.0022$$

$$\rightarrow C_4 = \frac{3.9383}{2.8772} = 1.3687, f(1.3687) = -0.0022$$

a	b	c	f(c)
1	1.5	1.3544	-0.3026
1.3544	1.5	1.3682	-0.0128
1.3682	1.5	1.3687	-0.0022
1.3687	1.5	1.3687	-0.0022

root $\rightarrow \boxed{1.3687} \text{ (Ans)}$

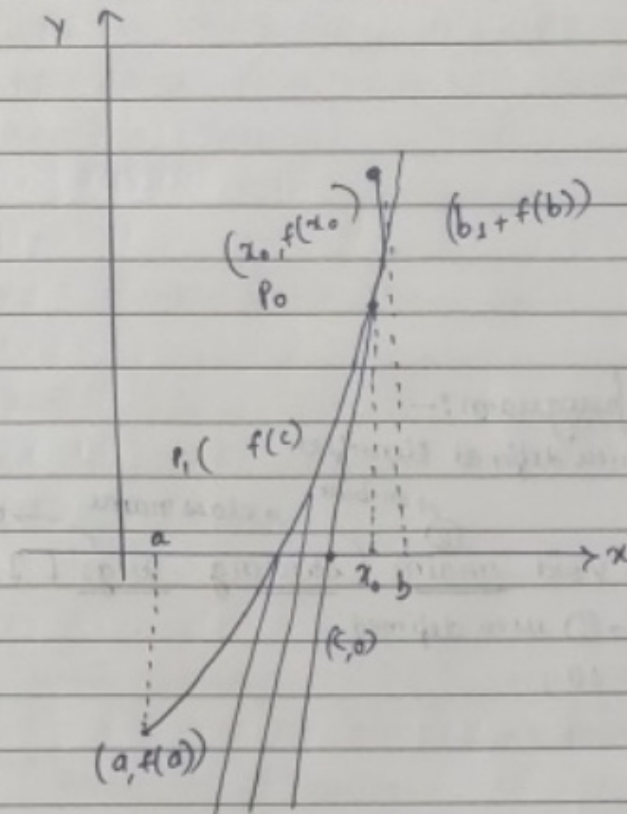
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③ Method $\rightarrow$ Newton Raphson Method :-

Rate of convergence = 2



Tangent at P_0 —

$$\rightarrow \frac{dy}{dx} = \frac{y - y_1}{x - x_1}$$

$$\rightarrow f'(x_0) = \frac{y - f(x_0)}{x - x_0}$$

$$\rightarrow f'(x_0) = \frac{0 - f(x_0)}{c - x_0}$$

$$\rightarrow c - x_0 = \frac{-f(x_0)}{f'(x_0)}$$

$$\rightarrow c = x_0 - \frac{f(x_0)}{f'(x_0)}, \quad (x, y) \text{ lies on tangent}$$

$$Q \rightarrow x^3 - 2x - 5 = 0$$

$$f'(x) = 3x^2 - 2$$

$$f(0) = -5$$

$$\Rightarrow 12 - 2 = 10$$

$$f(1) = -6$$

$$f(2) = -1$$

$$f(3) = 16$$

$$f(2.5) = 5.625$$

$\rightarrow$ Consider $x_0 = 2$

~~$$x_1 = 2 - \frac{f(2)}{f'(2)} = 2 - \frac{-1}{10} = 2.1$$~~

~~$$A(2.1) = 0$$~~

~~$$x_2 = 2.1 - \frac{f(2.1)}{f'(2.1)} = 2.1 - \frac{0.061}{11.23} = 2.0945$$~~

I ^{4th}

$$c_1 = \frac{2 + 1}{10} \Rightarrow \frac{20 + 1}{10} \Rightarrow \frac{21}{10} = 2.1$$

$$f(2.1) = 0.061$$

$$f'(2.1) = 11.23$$

II nd

$$x_0 = 2.1$$

$$c_2 = \frac{2.1 - 0.061}{11.23} \Rightarrow \frac{23.583 - 0.061}{11.23} \Rightarrow \frac{23.522}{11.23} \Rightarrow 2.0945$$

$$f(2.0945) = -0.0005$$

$$f'(2.0945) = 11.1607$$

III rd

?

Ques → Find the value of $\sqrt{12}$

Sol:-

$$x = \sqrt{12}$$

$$x^2 = 12$$

$$x^2 - 12 = 0$$

$$f(x) = x^2 - 12$$

$$f(0) = -12$$

$$f(1) = -11$$

$$f(2) = -8$$

$$f(3) = -3$$

$$f(4) = 4$$

$$f(5) = 13$$

$$f(3.5) = 0.25$$

→ consider $x_0 = 3.5$

Ist

$$C_1 = 3.5 - \frac{0.25}{7} \Rightarrow \frac{24.5 - 0.25}{7} \Rightarrow \frac{24.25}{7} \Rightarrow 3.4642$$

$$f(3.4642) = 0.0006$$

$$f'(3.4642) = 6.9284$$

IInd

$$\rightarrow C_2 = 3.4642 - \frac{0.0006}{6.9284} \Rightarrow \frac{24.0013 - 0.0006}{6.9284}$$

$$\Rightarrow \frac{24.0007}{6.9284} \Rightarrow 3.4641$$

$$f(3.4641) = -0.0000$$

$$f'(3.4641) = 6.9282$$

$$\text{Root} \rightarrow \boxed{3.4641}$$

Ques → $3x - \cos x = 1$

Sol: $f(x) = 3x - \cos x - 1$ $\Rightarrow f'(x) = 3 + \sin x$

$$f(0) = -2$$

$$f(1) = 1.4596$$

$$f(0.5) = -0.3775$$

→ consider $x_0 = 0.5$

$$C_1 = 0.5 + 0.3775 \Rightarrow 0.5 + 0.1084 \Rightarrow 0.6084$$

$$f(0.6084) \Rightarrow 0.0046$$

$$f'(0.6084) \Rightarrow 3.5715$$

→ $C_2 = 0.6084 - 0.0012$ $x_0 \Rightarrow 0.6084$
 $= 0.6072$

$$f(0.6072) = 0.0046$$

$$f'(0.6072) = 3.5705$$

→ $C_3 = 0.6072 - 0.0012$ $x_0 = 0.6072$
 $\Rightarrow 0.606$

$$f(0.606) = -3.932$$

$$f'(0.606) = 3.5695$$